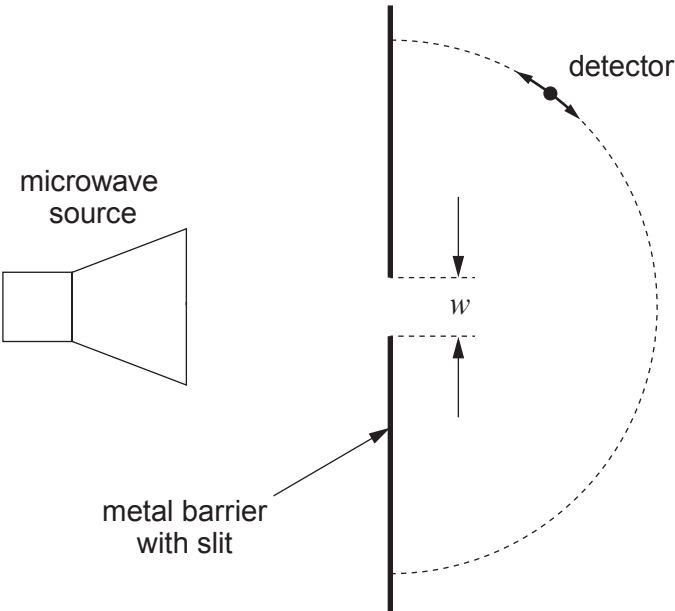


5. (a) In the set-up shown, a microwave detector moved along the semi-circular path detects diffracted waves at **all** points around the semicircle.



- (i) The width, w , of the gap is 25 mm. What can be deduced about the wavelength of the microwaves? [1]

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- (ii) Rhian suggests that making the gap wider will increase the intensity of the microwaves detected. Discuss to what extent she is correct. [2]

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Examiner
only

(b) In the set-up shown, slits S_1 and S_2 act as in-phase sources. Rhian detects **minima** of wave intensity at P and Q, and a single maximum between P and Q.

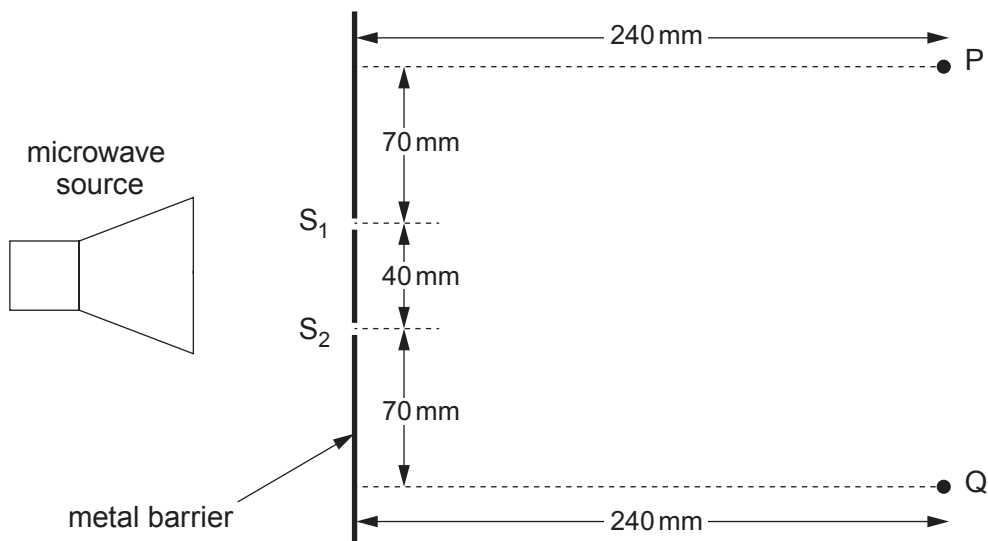


Diagram not to scale

Calculate the path lengths S_1P and S_2P , and **hence** the wavelength of the microwaves. [4]

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